Towards Controllable Speech Synthesis in the Era of Large Language Models: A Systematic Survey

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Abstract

Text-to-speech (TTS) has advanced from generating natural-sounding speech to enabling fine-grained control over attributes like emotion, timbre, and style. Driven by rising industrial demand and breakthroughs in deep learning, e.g., diffusion and large language models (LLMs), controllable TTS has become a rapidly growing research area. This survey provides **the first** comprehensive review of controllable TTS methods, from traditional control techniques to emerging approaches using natural language prompts. We categorize model architectures, control strategies, and feature representations, while also summarizing challenges, datasets, and evaluations in controllable TTS. This survey aims to guide researchers and practitioners by offering a clear taxonomy and highlighting future directions in this fast-evolving field. One can visit `https://github.com/imxtx/awesome-controllabe-speech-synthesis` for a comprehensive paper list and updates.

1 Introduction

Speech synthesis, also known as text-to-speech (TTS), aims to generate human-like speech from text (Dutoit, 1997), and has found broad applications in personal assistants (López et al., 2018), entertainment (Wang et al., 2019), and robotics (Marge et al., 2022). Recently, the success of large language models such as ChatGPT (OpenAI, 2022) has renewed interest in TTS for natural and intuitive human-computer interaction. Meanwhile, fine-grained control over speech attributes, such as emotion, timbre, and style, has become a key focus in both academia and industry, unlocking more expressive and personalized voice generation.

In the past decade, deep learning has driven remarkable advances in TTS, enabling high-quality synthesis (Tan et al., 2024; Ren et al., 2019; Du et al., 2024) and stronger control over speech attributes (Wang et al., 2018; Li et al., 2021; Zhou et al., 2024). Recent methods have expanded TTS to multi-modal inputs, including images (Rong and Liu, 2025) and videos (Choi et al., 2023). Meanwhile, the rise of LLMs (Zhao et al., 2023) has enabled controllable TTS guided by language prompts (Guo et al., 2023; Huang et al., 2024a), opening new possibilities for customized voice synthesis. Integrating TTS into LLMs has also gained extensive attention (Peng et al., 2024a). This rapid progress underscores the need for a comprehensive and timely survey to clarify current trends and guide future directions in controllable TTS.

While several surveys have examined parametric (Zen et al., 2009) and deep learning–based TTS (Triantafyllopoulos et al., 2023), they overlook TTS controllability and recent advances such as description–based methods (Guo et al., 2023; Yamamoto et al., 2024). The key differences between our survey and earlier work are: **1) Different Scope**: Klatt (1987) provided the first review of formant-based, concatenative, and articulatory TTS, with a strong focus on text analysis. Later, Tabet and Boughazi (2011); King (2014) explored statistics-based techniques. With the advent of neural networks, Ning et al. (2019); Tan et al. (2021b); Zhang et al. (2023a) surveyed neural model–based TTS, focusing on acoustics and vocoders. However, they rarely discuss the controllability. **2) Closer to Current Demands**: The need for controllable TTS is rapidly growing in industries like filmmaking, gaming, robotics, and virtual assistants. Yet, existing surveys rarely explore the gaps between current control techniques and real-world demands.

To fill this gap, we present the first comprehensive survey of emerging controllable TTS methods. We first define the core tasks (Sec. 2) and, as shown in Fig. 1, trace the evolution of methods across model architectures (Sec. 3.1), control strategies (Sec. 3.2), and feature representations (Sec. 3.3). We further summarize relevant datasets and evaluation metrics (Sec. 4), and discuss current

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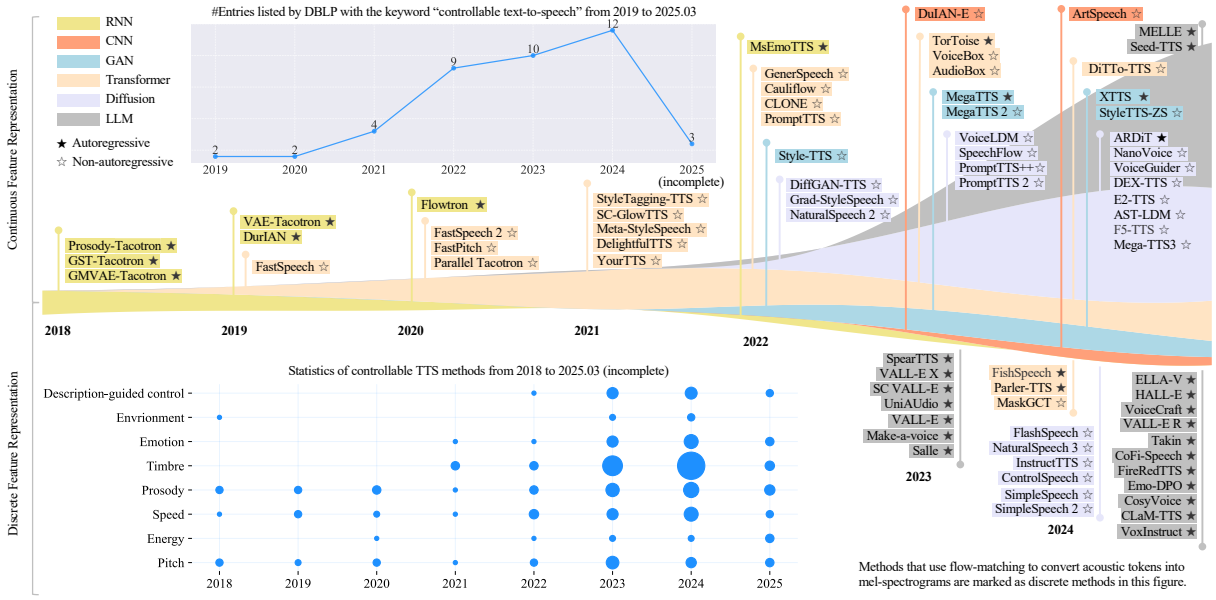


Figure 1: Recent trends in controllable TTS regarding architectures, feature representations, and control abilities.

challenges and future research directions (Sec. 5). For a history of controllable TTS and an overview of the TTS pipeline, see Appendices A.1 and A.2.

2 Main Tasks in Controllable TTS

Prosody Control is the most basic task in controllable TTS, aiming to manipulate low-level acoustic features such as pitch (Łańcucki, 2021), duration (Wang et al., 2025a), and energy (Chen et al., 2025). Prosody control ensures naturalness and expressiveness in TTS and is essential for rendering emphasis, rhythm, and nuance in speech.

Timbre Control aims to manipulate the acoustic characteristics that define voice quality (e.g., gender, age, nasality), enabling control over how a voice sounds beyond content and prosody. It supports personalized TTS (Du et al., 2024), voice conversion (Zhang et al., 2025b), and speaker identity editing (Huang et al., 2024a).

Emotion Control aims to enable the synthesis of emotional speech by manipulating the affective state of the generated voice (Kim et al., 2021). This improves human-computer interaction, storytelling (Rong et al., 2025b), and supports emotionally adaptive systems such as virtual assistants.

Style Control aims to control higher-level attributes of speech such as tone, formality, and discourse mode (e.g., newscast) (Zhou et al., 2024; Yang et al., 2024b). This is critical for adapting the speaking behavior of TTS systems to different contexts, audiences, and communication goals.

Language Control aims to enable TTS systems to synthesize speech in multiple languages (Zhang et al., 2023d), dialects (Di et al., 2024), or code-switched contexts (Chen et al., 2024d). It facilitates cross-lingual communication, multilingual agents, and regionally tailored speech applications.

Environment Control aims to simulate the acoustic characteristics of a specific setting, such as a park, office, or seaside, by conditioning synthesis on background noise and spatial cues (Lee et al., 2024; Kim et al., 2024b; Zhang et al., 2025c). Speech environment control is useful in filmmaking and audiobooks.

3 Methods in Controllable TTS

This section reviews controllable TTS from three perspectives: model architectures, feature representations, and control strategies, as shown in Fig. 1.

3.1 Model Architectures

The architectures of controllable TTS are primarily divided into two types, i.e., non-autoregressive (NAR) and autoregressive (AR) (See Table 1).

3.1.1 Non-Autoregressive Approaches

Non-autoregressive TTS models generate the entire output speech sequence $\mathbf{y} = (y_1, y_2, \ldots, y_T)$ in parallel given the input $\mathbf{x} = (x_1, x_2, \ldots, x_T)$:

$$\arg\max_{\theta} = P(\mathbf{y}|\mathbf{x}; \theta), \tag{1}$$

where θ denotes model parameters. In this part, we investigate the transformer, variational autoencoder (VAE), diffusion, and flow-based methods.

Method (Non-autoregressive)	Zero-shot	Controlability								Model Architectures		Feature	Release
		Pit.	Ene.	Spe.	Pro.	Tim.	Emo.	Env.	Des.	Acoustic Model	Vocoder		
FastSpeech (Ren et al., 2019)				✓	✓					Transformer	WaveGlow (Prenger et al., 2019)	MelS	2019.05
FastSpeech 2 (Ren et al., 2021a)		✓	✓	✓	✓					Transformer	Parallel WaveGAN (Yamamoto et al., 2020)	MelS	2020.06
FastPitch (Łańcucki, 2021)		✓			✓					Transformer	WaveGlow	MelS	2020.06
Parallel Tacotron (Elias et al., 2021a)					✓					Transformer + VAE + CNN	WaveRNN (Kalchbrenner et al., 2018)	MelS	2020.10
StyleTagging-TTS (Kim et al., 2021)	✓					✓	✓			Transformer + CNN	HiFi-GAN (Kong et al., 2020)	MelS	2021.04
SC-GlowTTS (Casanova et al., 2021)	✓					✓				Transformer + Flow	HiFi-GAN	MelS	2021.06
Meta-StyleSpeech (Min et al., 2021)	✓					✓				Transformer	MelGAN (Kumar et al., 2019)	MelS	2021.06
DelightfulTTS (Liu et al., 2021)		✓		✓	✓					Transformer + CNN	HiFiNet (Liu et al., 2021)	MelS	2021.11
YourTTS (Casanova et al., 2022)	✓					✓				Transformer + Flow	HiFi-GAN	LinS	2021.12
StyleTTS (Li et al., 2025b)	✓					✓				CNN + RNN	HiFi-GAN	MelS	2022.05
GenerSpeech (Huang et al., 2022b)	✓					✓				Transformer + Flow	HiFi-GAN	MelS	2022.05
Cauliflow (Abbas et al., 2022)				✓	✓					BERT + Flow	UP WaveNet (Jiao et al., 2021)	MelS	2022.06
CLONE (Liu et al., 2022)		✓		✓	✓					Transformer + CNN	WaveNet (Van Den Oord et al., 2016)	MelS + LinS	2022.07
PromptTTS (Guo et al., 2023)		✓	✓	✓	✓	✓	✓		✓	BERT + Transformer	HiFi-GAN	MelS	2022.11
Grad-StyleSpeech (Kang et al., 2023)	✓					✓				Score-based Diffusion	HiFi-GAN	MelS	2022.11
NaturalSpeech 2 (Shen et al., 2024)	✓					✓				Diffusion	RVQ-based (Shen et al., 2024)	Latent Feature	2023.04
PromptStyle (Liu et al., 2023a)	✓	✓			✓	✓	✓		✓	VITS + Flow	HiFi-GAN	MelS	2023.05
StyleTTS 2 (Li et al., 2023)	✓				✓	✓	✓			Flow-based Diffusion + GAN	HifiGAN / iSTFTNet (Kaneko et al., 2022)	MelS	2023.06
VoiceBox (Le et al., 2024)	✓					✓				Transformer + Flow	HiFi-GAN	MelS	2023.06
MegaTTS 2 (Jiang et al., 2024)	✓				✓	✓	✓			Decoder-only Transformer + GAN	HiFi-GAN	MelS	2023.07
PromptTTS 2 (Leng et al., 2023)		✓	✓	✓	✓	✓			✓	Diffusion	RVQ-based (Leng et al., 2023)	Latent Feature	2023.09
VoiceLDM (Lee et al., 2024)		✓			✓	✓	✓	✓	✓	Diffusion	HiFi-GAN	MelS	2023.09
DurIAN-E (Gu et al., 2023)		✓		✓	✓					CNN + RNN	HiFi-GAN	MelS	2023.09
PromptTTS++ (Shimizu et al., 2024)		✓		✓	✓	✓	✓		✓	Transformer + Diffusion	BigVGAN (1l Lee et al., 2023)	MelS	2023.09
SpeechFlow (Liu et al., 2024a)	✓					✓				Transformer + Flow	HiFi-GAN	MelS	2023.10
P-Flow (Kim et al., 2024c)	✓					✓				Transformer + Flow	HiFi-GAN	MelS	2023.10
E3 TTS (Gao et al., 2023)	✓					✓				Diffusion	Not required	Waveform	2023.11
HierSpeech++ (Lee et al., 2023b)	✓					✓				Transformer + VAE + Flow	BigVGAN	MelS	2023.11
Audiobox (Vyas et al., 2023)	✓	✓		✓	✓	✓		✓	✓	Transformer + Flow	HiFi-GAN	MelS	2023.12
FlashSpeech (Ye et al., 2024)	✓					✓				Latent Consistency Model	EnCodec	Token	2024.04
NaturalSpeech 3 (Ju et al., 2024)	✓			✓	✓	✓				Transformer + Diffusion	FACodec (Ju et al., 2024)	Token	2024.04
InstructTTS (Yang et al., 2024b)		✓		✓	✓	✓	✓		✓	Transformer + Diffusion	HiFi-GAN	Token	2024.05
ControlSpeech (Ji et al., 2024c)	✓	✓	✓	✓	✓	✓	✓		✓	Transformer + Diffusion	FACodec	Token	2024.06
AST-LDM (Kim et al., 2024b)						✓		✓	✓	Diffusion + VAE	HiFi-GAN	MelS	2024.06
SimpleSpeech (Yang et al., 2024c)	✓					✓				Transformer + Diffusion	SQ Codec (Yang et al., 2024c)	Token	2024.06
DiTTo-TTS (Lee et al., 2025)	✓			✓		✓				DiT + VAE	BigVGAN	MelS	2024.06
E2 TTS (Eskimez et al., 2024)	✓					✓				Transformer + Flow	BigVGAN	MelS	2024.06
MobileSpeech (Ji et al., 2024a)	✓					✓				Transformer	Vocos (Siuzdak, 2024)	Token	2024.06
DEX-TTS (Park et al., 2024a)	✓					✓				Diffusion	HiFi-GAN	MelS	2024.06
ArtSpeech (Wang et al., 2024)	✓					✓				RNN + CNN	HiFi-GAN	MelS + Energy + F0	2024.07
CCSP (Xiao et al., 2024)	✓					✓				Diffusion	RVQ-based (Xiao et al., 2024)	Token	2024.07
SimpleSpeech 2 (Yang et al., 2024a)	✓			✓		✓				Flow-based DiT	SQ Codec	Token	2024.08
E1 TTS (Liu et al., 2025b)	✓					✓				DiT + Flow	BigVGAN	Token + MelS	2024.09
StyleTTS-ZS (Li et al., 2024)	✓					✓				Flow-based Diffusion + GAN	Mel-based Decoder (Li et al., 2024)	MelS	2024.09
NansyTTS (Yamamoto et al., 2024)	✓	✓		✓	✓	✓			✓	Transformer	NANSY++ (Yamamoto et al., 2024)	MelS	2024.09
NanoVoice (Park et al., 2024b)	✓					✓				Diffusion	BigVGAN	MelS	2024.09
MS^{2}KU-VTTS (He et al., 2024)								✓	✓	Transformer	BigVGAN	MelS	2024.10
MaskGCT (Wang et al., 2025b)	✓			✓		✓				Transformer + Flow	Vocos	Token	2024.10
EmoSphere++ (Cho et al., 2024)	✓				✓	✓	✓			Transformer + Flow	BigVGAN	MelS	2024.11
EmoDubber (Cong et al., 2024)	✓				✓	✓	✓			Transformer + Flow	Flow-based (Cong et al., 2024)	MelS	2024.12
HED (Inoue et al., 2024)	✓						✓			Flow-based Diffusion	Vocos	MelS	2024.12
DiffStyleTTS (Liu et al., 2025a)		✓	✓	✓	✓	✓				Transformer + Diffusion	HiFi-GAN	MelS	2025.01
DrawSpeech (Chen et al., 2025)			✓		✓					Diffusion	HiFi-GAN	MelS	2025.01
ProEmo (Zhang et al., 2025a)		✓	✓				✓		✓	Transformer	HiFi-GAN	MelS	2025.01

Method (Autoregressive)	Zero-shot	Controlability								Model Architectures		Feature	Release
		Pit.	Ene.	Spe.	Pro.	Tim.	Emo.	Env.	Des.	Acoustic Model	Vocoder		
Prosody-Tacotron (Skerry-Ryan et al., 2018)		✓			✓					RNN	WaveNet	MelS	2018.03
GST-Tacotron (Stanton et al., 2018)		✓			✓					CNN + RNN	Griffin-Lim	LinS	2018.03
GMVAE-Tacotron (Hsu et al., 2018)		✓		✓	✓			✓		VAE	WaveRNN	MelS	2018.12
VAE-Tacotron (Zhang et al., 2019)		✓		✓	✓					VAE + CNN + RNN	WaveNet	MelS	2019.02
DurIAN (Yu et al., 2020)		✓		✓	✓					CNN + RNN	Multi-band WaveRNN (Yu et al., 2020)	MelS	2019.09
Flowtron (Valle et al., 2020)		✓		✓	✓					CNN + RNN	WaveGlow	MelS	2020.07
MsEmoTTS (Lei et al., 2022)		✓			✓		✓			CNN + RNN	WaveRNN	MelS	2022.01
VALL-E (Wang et al., 2023a)	✓					✓				Decoder-only Transformer	EnCodec	Token	2023.01
SpearTTS (Kharitonov et al., 2023)	✓					✓				Decoder-only Transformer	SoundStream (Zeghidour et al., 2021)	Token	2023.02
VALL-E X (Zhang et al., 2023d)	✓					✓				Decoder-only Transformer	EnCodec	Token	2023.03
Make-A-Voice (Huang et al., 2023b)	✓					✓				Encoder-decoder Transformer	Unit-based (Huang et al., 2023b)	Token	2023.05
TorToise (Betker, 2023)						✓				Decoder-only Transformer + Diffusion	UnivNet (Jang et al., 2021)	MelS	2023.05
MegaTTS (Jiang et al., 2023)	✓					✓				Decoder-only Transformer + GAN	HiFi-GAN	MelS	2023.06
SC VALL-E (Kim et al., 2023)	✓	✓		✓	✓	✓	✓			Decoder-only Transformer	EnCodec	Token	2023.07
Salle (Ji et al., 2024b)		✓	✓	✓	✓	✓	✓		✓	Decoder-only Transformer	EnCodec	Token	2023.08
UniAudio (Yang et al., 2023b)	✓	✓		✓	✓	✓			✓	Decoder-only Transformer	EnCodec	Token	2023.10
ELLA-V (Song et al., 2024)	✓					✓				Decoder-only Transformer	EnCodec	Token	2024.01
Base TTS (Łajszczak et al., 2024)	✓					✓				Decoder-only Transformer	HiFi-GAN + BigVGAN	Token	2024.02
CLaM-TTS (Kim et al., 2024a)	✓					✓				Encoder-decoder Transformer	BigVGAN	Token + MelS	2024.04
RALL-E (Xin et al., 2024)	✓					✓				Decoder-only Transformer	SoundStream	Token	2024.05
ARDiT (Liu et al., 2024b)	✓			✓		✓				Decoder-only DiT	BigVGAN	MelS	2024.06
VALL-E R (Han et al., 2024)	✓					✓				Decoder-only Transformer	Vocos	Token	2024.06
VALL-E 2 (Chen et al., 2024a)	✓					✓				Decoder-only Transformer	Vocos	Token	2024.06
Seed-TTS (Anastassiou et al., 2024)	✓					✓	✓			Decoder-only Transformer + DiT	Unknown	Latent Feature	2024.06
VoiceCraft (Peng et al., 2024b)	✓					✓				Decoder-only Transformer	HiFi-GAN	Token	2024.06
XTTS (Casanova et al., 2024)	✓					✓				Decoder-only Transformer	HiFi-GAN-based (Casanova et al., 2024)	Token + MelS	2024.06
CosyVoice (Du et al., 2024)	✓	✓		✓	✓	✓	✓		✓	Decoder-only Transformer + Flow	HiFi-GAN	Token	2024.07
MELLE (Meng et al., 2024)	✓					✓				Decoder-only Transformer	HiFi-GAN	MelS	2024.07
VoxInstruct (Zhou et al., 2024)	✓	✓	✓	✓	✓	✓	✓		✓	Decoder-only Transformer	Vocos	Token	2024.08
Emo-DPO (Gao et al., 2024)							✓			Decoder-only Transformer	HiFi-GAN	Token + MelS	2024.09
FireRedTTS (Guo et al., 2024a)	✓				✓	✓				Decoder-only Transformer + Flow	BigVGAN	Token + MelS	2024.09
CoFi-Speech (Guo et al., 2024b)	✓					✓				Decoder-only Transformer	BigVGAN	Token + MelS	2024.09
Takin (Chen et al., 2024b)	✓	✓		✓	✓	✓	✓		✓	Decoder-only Transformer + Flow	HiFi-GAN	Token + MelS	2024.09
HALL-E (Nishimura et al., 2024)	✓					✓				Decoder-only Transformer	EnCodec	Token	2024.10
FishSpeech (Liao et al., 2024)	✓					✓				Decoder-only Transformer	Firefly-GAN (Liao et al., 2024)	Token	2024.11
SLAM-Omni (Chen et al., 2024c)	✓				✓	✓				Decoder-only Transformer	HiFi-GAN	Token + MelS	2024.12
IST-LM (Yang et al., 2024e)	✓				✓	✓				Decoder-only Transformer	HiFi-GAN	Token + MelS	2024.12
KALL-E (Zhu et al., 2024)	✓				✓	✓	✓			Decoder-only Transformer	WaveVAE (Zhu et al., 2024)	Latent Feature	2024.12
IDEA-TTS (Lu et al., 2025)	✓					✓		✓		Transformer	Flow-based (Lu et al., 2025)	LinS + MelS	2024.12
FleSpeech (Li et al., 2025a)	✓	✓	✓	✓	✓	✓	✓		✓	Flow-based DiT	WaveGAN (Donahue et al., 2018)	Latent Feature	2025.01
Step-Audio (Huang et al., 2025)	✓				✓	✓	✓		✓	Decoder-only Transformer	Flow-based (Huang et al., 2025)	Token	2025.02
Vevo (Zhang et al., 2025b)	✓				✓	✓	✓			Decoder-only Transformer	BigVGAN	Token + MelS	2025.02
Spark-TTS (Wang et al., 2025a)	✓	✓		✓	✓	✓				Decoder-only Transformer	BiCodec (Wang et al., 2025a)	Token	2025.03
EmoVoice (Yang et al., 2025)							✓		✓	Decoder-only Transformer	HiFi-GAN	Token	2025.04

Abbreviations: Pit(ch), Ene(rgy), Spe(ed), Pro(sody), Tim(bre), Emo(tion), Env(ironment), Des(cription). MelS: Mel Spectrogram. LinS: Linear Spectrogram.

Table 1: A summary of existing controllable neural-based methods.

Transformer-based Methods. Transformers enable efficient context modeling and parallel TTS. FastSpeech (Ren et al., 2019) introduced a non-autoregressive transformer that improves inference speed and stability via duration prediction. FastSpeech 2 (Ren et al., 2021a) adds pitch and energy control, removing the need for distillation and boosting voice quality. FastPitch (Łańcucki, 2021) further incorporates direct pitch prediction into its architecture, enabling pitch manipulation.

VAE-based Methods. VAEs enable structured, continuous latent representations by optimizing a variational lower bound. VAEs have been leveraged to enhance prosody, emotion, and style control. Hsu et al. (2018) proposed a hierarchical VAE to control noise and speaking rate. Zhang et al. (2019) introduced disentangled VAE representations for effective prosody and emotion transfer. Parallel Tacotron (Elias et al., 2021b) uses a VAE-based residual encoder with iterative spectrogram loss to improve speech naturalness. CLONE (Liu et al., 2022) further improves prosody and energy modeling using conditional VAEs with normalizing flows (Kobyzev et al., 2020) and adversarial training, achieving state-of-the-art quality and control. These advances underscore VAEs' versatility in expressive and controllable speech synthesis.

Diffusion-based Methods. Diffusion mod-

els (Ho et al., 2020) generate speech by reversing a noise injection process: noise is gradually added during the forward phase and removed in the reverse phase to synthesize high-quality audio. NaturalSpeech 2 (Shen et al., 2024) uses a latent diffusion-based codec with quantized latent vectors, while NaturalSpeech 3 (Ju et al., 2024) decomposes speech into independent attribute subspaces with factorized diffusion-based codecs. DEX-TTS (Park et al., 2024a) improves diffusion transformer (DiT)-based networks via overlapping patches and frequency-aware embeddings. E3 TTS (Gao et al., 2023) eliminates intermediate features by directly modeling waveforms through diffusion. Text-to-audio models such as AudioLDM (Liu et al., 2023b) and Make-An-Audio (Huang et al., 2023a) can also generate speech using latent diffusion models.

Flow-based Methods. Flow models use invertible flows (Rezende and Mohamed, 2015; Lipman et al., 2023) to map speech features to simple distributions, typically Gaussians (Prenger et al., 2019), enabling direct, high-fidelity generation via inversion. Recent models adopt flow-matching (Lipman et al., 2023) for efficient, non-autoregressive synthesis: Audiobox (Vyas et al., 2023), P-Flow (Kim et al., 2024c), and VoiceBox (Le et al., 2024) consider TTS as speech infilling tasks, predicting masked mel-spectrograms. FlashSpeech (Ye et al., 2024) trains a latent consistency model using adversarial training, achieving one- or two-step synthesis. Inspired by audio infilling, E2 TTS (Eskimez et al., 2024) uses filler-augmented text sequences to generate mel-spectrograms with human-level quality. F5-TTS (Chen et al., 2024d) builds on this with ConvNeXt v2 (Woo et al., 2023) to enhance text-speech alignment by directly learning flows conditioned on text and reference speech. E1 TTS (Liu et al., 2025b) distills rectified flow-based diffusion models (Liu et al., 2023c) into one-step generators via distribution matching, reducing sampling cost.

3.1.2 Autoregressive Approaches

Autoregressive TTS models predict the speech sequence $\mathbf{y} = (y_1, \ldots, y_T)$ given input $\mathbf{x}$ as:

$$\arg\max_{\theta} = \prod_{t=1}^{T} P(y_t | y_{<t}, \mathbf{x}; \theta), \qquad (2)$$

where each frame y_t depends on all previous outputs $y_{<t}$ and the transcript $\mathbf{x}$. While this enables effective modeling of implicit duration and long-range context, autoregressive TTS models suffer from slower inference, making them more suitable for applications where flexibility is prioritized over speed. This part investigates recurrent neural networks (RNN) and LLM-based methods.

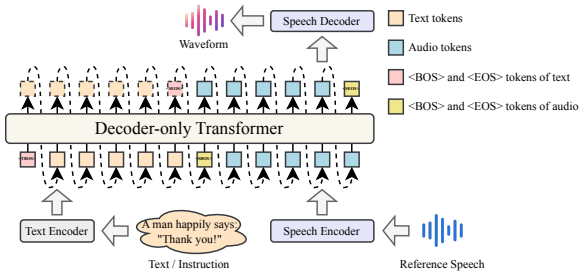


Figure 2: The typical architecture of LLM-based TTS.

RNN-based Methods. RNNs enable natural-sounding speech synthesis with adjustable prosody, pitch, and emotion. Prosody-Tacotron (Skerry-Ryan et al., 2018) extends Tacotron (Wang et al., 2017) by introducing explicit prosodic controls. Wang et al. (2018) proposed Global Style Tokens (GST), enabling unsupervised style transfer. Emotion-controllable models such as Li et al. (2021) introduced emotion embeddings and style alignment to modulate emotional intensity. MsEmoTTS (Lei et al., 2022) refined this with a hierarchical structure capturing emotion at global, utterance, and local levels, enabling more nuanced synthesis. These developments bring synthetic speech significantly closer to human expressiveness.

LLM-based Methods. LLM-based TTS is inspired by the success of in-context learning in LLMs. As illustrated in Fig. 2, these approaches typically input target text or instructions with an optional reference speech, using autoregressive decoder-only transformers to generate speech tokens or features, which are then decoded into waveforms. VALL-E (Wang et al., 2023a) pioneered LLM-based zero-shot TTS by framing it as a conditional language modeling task. It uses EnCodec (Défossez et al., 2023a) to discretize waveforms into tokens and adopts a two-stage pipeline: an autoregressive model generates coarse audio tokens, followed by a non-autoregressive model for iterative refinement. This hierarchical modeling of semantic and acoustic tokens has laid the groundwork for many subsequent methods, such as VALL-E X (Zhang et al., 2023d), ELLA-V (Song et al., 2024), RALL-E (Xin et al., 2024), VALL-E R (Han et al., 2024), MELLE (Meng et al., 2024), and HALL-E (Nishimura et al., 2024). Beyond the VALL-E series, recent work has further im-

proved text-speech alignment, quality, and robustness. SpearTTS (Kharitonov et al., 2023) and Make-a-Voice (Huang et al., 2023b) leverage semantic tokens to better bridge text and acoustic features. FireRedTTS (Guo et al., 2024a) refines the tokenizer architecture for improved reconstruction quality. CoFi-Speech (Guo et al., 2024b) adopts a coarse-to-fine, multi-scale generation strategy to produce natural, intelligible speech.

3.1.3 Research Trend

Traditional CNN/RNN TTS models face inherent constraints. RNNs (e.g., Tacotron) are slow due to autoregressive inference and struggle with long-term dependencies. CNNs (e.g., Deep Voice) lack global prosody modeling. Both require explicit feature engineering for attributes like emotion and often trade synthesis quality for efficiency (e.g., WaveNet's high fidelity but high latency). **Flow-based models** (e.g., Matcha-TTS, F5-TTS) enable non-autoregressive, parallel synthesis with probabilistic control over acoustic features, improving speed and flexibility but increasing training complexity and dataset requirements. LLM-based models (e.g., VALL-E, InstructTTS) offer natural language-driven control and zero-shot voice cloning, supporting context-aware synthesis, but suffer from high computational cost and potential acoustic artifacts from discrete tokenization. **Hybrid architectures** (e.g., CosyVoice) integrate LLM-guided semantic conditioning into flow-based generators, combining high-fidelity synthesis with intuitive, instruction-based control. Users can specify attributes like emotion or speaking style in natural language without compromising audio quality. Future controllable TTS should balance efficiency, fidelity, and expressiveness, generalizing across voices, styles, and languages. Bridging instruction-based control and acoustic precision remains a key challenge, motivating advances in modular architectures, instruction grounding, and speech-text-instruction alignment. Fig. 3 summarizes the evolution and future direction of TTS model architectures.

3.2 Control Strategies

As illustrated in Fig. 4, control strategies in TTS can be broadly categorized into four types: style tagging, reference speech prompt, natural language descriptions, and instruction-guided control.

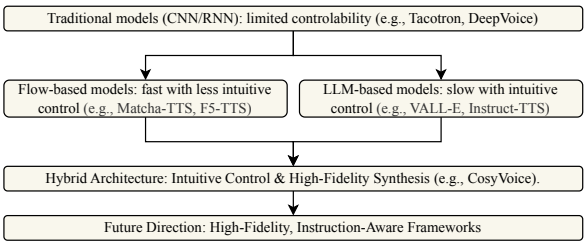


Figure 3: The evolution of TTS model architectures

3.2.1 Style Tagging

This paradigm enables the adjustment of key attributes such as pitch, energy, speech rate, and emotion, which can be controlled using either categorical labels or continuous values. "Tagging" refers to using a control signal to control a specific speech attribute. 1) Some approaches use *discrete labels* to control speech attributes. For example, StyleTagging-TTS (Kim et al., 2021) denotes speech styles with short phrases or words (e.g., angry, happy), learning the relationship between linguistic and style embeddings. Emo-DPO (Gao et al., 2024) enables emotion control through Direct Preference Optimization (DPO) (Rafailov et al., 2023) with LLMs. Spark-TTS (Wang et al., 2025a) provides coarse and fine-grained control, allowing pitch and speaking rate modifications via specially designed tokens and reference speech. 2) Other methods adjust *continuous input signals*. DiffStyleTTS (Liu et al., 2025a) models prosody hierarchically, enabling control over pitch, energy, duration, and style via guiding scale factors. DrawSpeech (Chen et al., 2025) lets users sketch prosody contours, which are refined and converted into detailed speech by a diffusion model, offering control over intonation. 3) Speech attributes can be controlled by *modifying latent features*. Cauliflow (Abbas et al., 2022) adjusts speech rate and pausing through a flow-based model conditioned on user-defined parameters. DiTTo-TTS (Lee et al., 2025) uses a DiT to control speech rate by modifying latent length predictions. These methods show great potential in controlling speech attributes by adjusting input signals or latent variables. However, these methods are limited in expressive diversity, as they can only model a small set of pre-defined attributes.

3.2.2 Reference Speech Prompt

This paradigm aims to customize the synthesized voice using only a few seconds of reference speech. Similar to LLM-based methods, it takes both text

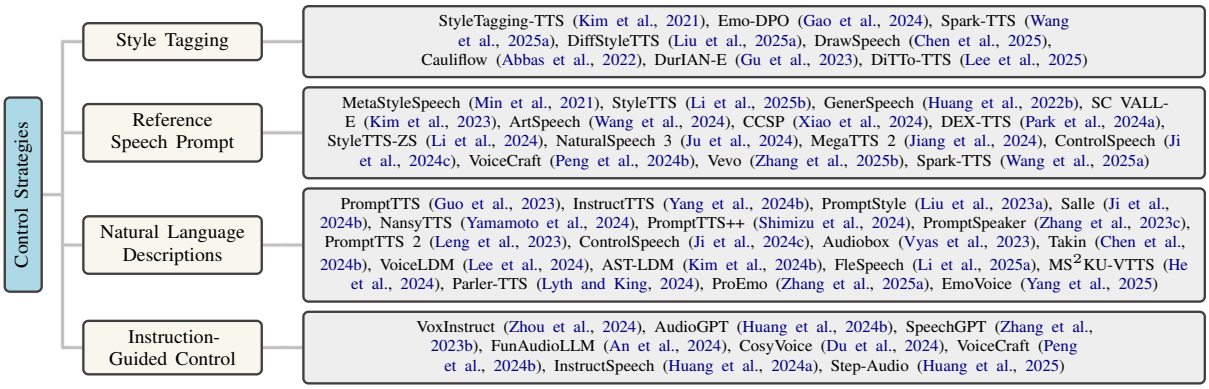


Figure 4: A taxonomy of controllable TTS from the perspective of control strategies.

and reference speech as input to a conditional TTS model, which generates speech based on both semantic and acoustic features. MetaStyleSpeech (Min et al., 2021) employs adaptive normalization for style conditioning, enabling robust zero-shot performance. GenerSpeech (Huang et al., 2022b) introduces a multilevel style adapter for improved zero-shot style transfer to out-of-domain custom voices. SC VALL-E (Kim et al., 2023) integrates style tokens and scale factors for controlling emotion, speaking style, and other acoustic features in the generated speech. DEX-TTS (Park et al., 2024a) separates time-invariant and time-variant style components, allowing the extraction of diverse styles. StyleTTS-ZS (Li et al., 2024) uses distilled time-varying style diffusion to capture varied speaker identities and prosodies. MegaTTS 2 (Jiang et al., 2024) introduces an acoustic autoencoder to separate prosody and timbre in the latent space, enabling style transfer to any timbre. ControlSpeech (Ji et al., 2024c) employs bidirectional attention and parallel decoding to control timbre, style, and content in a zero-shot manner.

3.2.3 Natural Language Descriptions

Recent studies have explored controlling speech attributes using natural language descriptions, offering better user-friendliness. PromptTTS (Guo et al., 2023) uses manually annotated prompts to describe five key speech attributes. InstructTTS (Yang et al., 2024b) introduces a three-stage training procedure to extract semantics from natural language prompts. NansyTTS (Yamamoto et al., 2024) enables cross-lingual control by pairing a TTS model with a description controller trained on a different language using shared timbre and style representations. To address the limitations of textual prompts in capturing speaker characteristics, PromptTTS++ (Shimizu et al., 2024) enhances prompt richness by using additional speaker description prompts. PromptTTS 2 (Leng et al., 2023) introduces a variation network to model residual variability beyond the prompt. Further efforts extend controllability to the environmental context. VoiceLDM (Lee et al., 2024) and AST-LDM (Kim et al., 2024b) extend AudioLDM (Liu et al., 2023b) by incorporating content prompts to enable environmental conditioning. MS^{2}KU-VTTS (He et al., 2024) further enhances environmental perception by mixing environmental images into the prompt, enabling more immersive speech generation.

3.2.4 Instruction-Guided Synthesis

Description-based TTS methods separate inputs into content and description prompts, diverging from the unified instruction formats used in chatbots. To address this, VoxInstruct (Zhou et al., 2024) reframes TTS as a general instruction-to-speech task, where a single natural language prompt conveys both content and style descriptions. CosyVoice (Du et al., 2024) enhances this paradigm using supervised semantic tokens derived from ASR models. It combines LLM-driven token generation with flow-matching synthesis, enabling precise control over speaker identity, emotion, pitch, speed, and paralinguistic cues through natural language instructions. AudioGPT (Huang et al., 2024b) is a multimodal LLM-based agent, incorporating multiple modules for speech understanding, synthesis, and style conversion. StepAudio (Huang et al., 2025) introduces a speech-text model with an instruction-driven TTS module, enabling dynamic control over dialects, emotions, singing, rapping, and speaking styles. These advancements push toward more intuitive, instruction-driven speech generation.

3.2.5 Instruction-Guided Editing

Some methods also support speech editing via user instructions. VoiceCraft (Peng et al., 2024b) uses a decoder-only transformer with causal masking and delayed stacking for bidirectional, context-aware instruction-guided editing, such as insertion, deletion, and substitution, while maintaining high naturalness. InstructSpeech (Huang et al., 2024a) trains a multi-task LLM on <instruction, input, output> triplets with task embeddings and hierarchical adapters, allowing content and acoustic attributes control. It supports flexible, free-form speech editing and task adaptation by multi-step reasoning.

3.2.6 Research Trend

The evolution of TTS control has moved from basic attribute manipulation to sophisticated, instruction-guided synthesis, reflecting AI's trend toward intuitive, fine-grained control. Early methods like **style tagging** controlled predefined attributes (e.g., pitch, emotion) but offered limited expressive diversity. **Reference speech prompts** enabled zero-shot TTS and voice cloning, separating timbre from style for greater personalization. To improve user-friendliness, **natural language descriptions** (e.g., PromptTTS) allowed users to specify vocal characteristics through text. The latest advance, **instruction-guided control**, leverages LLMs to interpret free-form instructions combining content and style. Systems like VoxInstruct and CosyVoice generate nuanced speech, including paralinguistic sounds, enabling highly precise, user-centric synthesis. Overall, the progression from tags to natural instructions shows a clear trajectory toward more **expressive, personalized, and intuitive TTS**, driven by LLM integration. Table 2 in the Appendix provides a summary of the strengths and weaknesses of each control strategy.

3.3 Feature Representations

The learning and choice of feature representations critically affect flexibility, naturalness, and controllability. This subsection discusses speech attribute disentanglement and compares continuous and discrete representations, highlighting their trade-offs.

Speech Attribute Disentanglement. Attribute disentanglement aims to isolate distinct speech factors, such as speaker identity, emotion, prosody, and content, into separate latent representations. The two main approaches are: 1) *Adversarial training* (Goodfellow et al., 2020) uses auxiliary classifiers to penalize the presence of unwanted attributes in a latent space. An encoder learns to "fool" these classifiers, resulting in representations that are invariant to specific attributes like speaker (Yang et al., 2021; Hsu et al., 2019; Lee et al., 2021), emotion (Li et al., 2022), and style (Li et al., 2023). 2) *Information bottleneck* uses small-capacity or independent encoder branches to isolate attributes. Each branch encodes one factor (e.g., content, prosody) (Ju et al., 2024), often with adversarial or reconstruction losses to discourage leakage of other information. These methods are often combined. Regularization via KL divergence (Lu et al., 2023) or quantization (Zhang et al., 2025b) also plays a key role in enforcing disentanglement.

Continuous Representations. Continuous representations model speech in a continuous feature space, preserving acoustic details. The key advantages are: 1) Fine-grained detail retention for natural and expressive synthesis; 2) Inherent encoding of prosody, pitch, and emotion, aiding controllable and emotional TTS; 3) Enable smooth audio reconstruction without quantization artifacts. GAN-based (Kong et al., 2020; Yamamoto et al., 2020), VAE-based (Lee et al., 2023b, 2025), flow-based (Kim et al., 2024d; Casanova et al., 2022), and diffusion-based methods (Kong et al., 2021; Huang et al., 2022a) often utilize continuous feature representations. However, they are computationally intensive and demand large models and datasets for high-fidelity audio generation.

Discrete Tokens. Discrete token-based TTS uses quantized units or phoneme-like tokens as acoustic features, which are often derived from quantization (Zeghidour et al., 2021) or learned embeddings (Hsu et al., 2021). The advantages of discrete tokens are: 1) Discrete tokens can encode phonemes or sub-word units, making them concise and computationally efficient. 2) Discrete tokens often allow TTS systems to require fewer samples to learn and generalize, compared with continuous features. 3) Using discrete tokens simplifies cross-modal TTS applications like description-based TTS, as they are suitable for LLM training. LLM-based methods (Zhou et al., 2024; Yang et al., 2024b; Du et al., 2024) often adopt discrete tokens as acoustic features. However, discrete feature learning may cause information loss or lack the nuanced details in continuous features.

Table 1 summarizes the features used in existing methods. We also compare speech quantization and tokenization in Appendix A.3 and summarize open-source methods in Table 5 in the Appendix.

4 Datasets and Evaluation Methods

4.1 Datasets

Fully controllable TTS systems require large, diverse, and finely annotated datasets to generate expressive, attribute-controllable speech. There are mainly three types of datasets for controllable TTS:

Tag-based Datasets. Tag-based datasets contain speech recordings annotated with predefined discrete attribute labels that describe various aspects of the speech audio (Zhou et al., 2022; Busso et al., 2008; Ringeval et al., 2013; Bagher Zadeh et al., 2018). Common attributes include pitch, energy, speaking rate, age, gender, emotion, emphasis, accent, and topic. By leveraging attribute labels, models can dynamically adjust specific speech characteristics, enabling more expressive synthesis.

Description-based Datasets. Description-based datasets pair speech samples with rich, free-form textual descriptions that capture nuanced attributes such as intonation, prosody, speaking style, and emotional tone (Guo et al., 2023; Ji et al., 2024b; Jin et al., 2024; Lyth and King, 2024). Unlike tag-based datasets with predefined categorical labels, these datasets allow models to interpret and generate speech from natural language prompts, enabling context-aware and highly expressive synthesis.

Dialogue Datasets. Dialogue datasets (Byrne et al., 2019; Lee et al., 2023a; Yang et al., 2022) contain multi-turn conversational speech involving two or more speakers, emphasizing natural interaction features such as turn-taking, contextual dependencies, speaker intent, pauses, and prosodic variation. These datasets are essential for generating dynamic and contextually appropriate speech for interactive systems.

By leveraging these datasets, researchers can develop more expressive, context-aware, and highly controllable TTS models. Table 3 in the Appendix summarizes publicly available datasets.

4.2 Evaluation Methods

4.2.1 Objective and Subjective Metrics

Objective Metrics. Objective metrics enable automated and reproducible evaluation. *Mel Cepstral Distortion* (MCD) (Kominek et al., 2008) quantifies spectral distance between synthesized and reference speech. MCD below 4 suggests good synthesis, while values above 6 imply distortion. *Fréchet DeepSpeech Distance* (FDSD) (Bińkowski et al., 2020) evaluates speech quality by measuring the distributional distance between synthesized and real speech in the embedding space of a pre-trained speech recognition model, such as DeepSpeech (Hannun et al., 2014). It compares the mean and covariance of extracted features; thus, a lower FDSD indicates higher perceptual similarity. *Word Error Rate* (WER) (Wikipedia, 2024) quantifies speech intelligibility by comparing recognized and reference transcripts. *Cosine Similarity* assesses speaker similarity by comparing speaker embeddings (extracted using models like ECAPA-TDNN (Desplanques et al., 2020) or x-vectors (Snyder et al., 2018)) of synthesized and reference speech. Higher values indicate better voice cloning. *Perceptual Evaluation of Speech Quality* (PESQ) (Rix et al., 2001) evaluates the intelligibility and distortion of synthesized speech by modeling human auditory perception.

Subjective Metrics. Subjective metrics assess the perceptual quality of synthesized speech based on human judgments, capturing aspects like expressiveness and style similarity. *Mean Opinion Score* (MOS) rates synthesized speech (e.g., naturalness) on a 1–5 scale. Though effective in capturing human perception, MOS is costly for large-scale use. *Comparison MOS* (CMOS) (Loizou, 2011) assesses relative quality by asking participants compare paired samples. Both are averaged across listeners. *AB/ABX Tests* present listeners with two samples (AB) by different methods or two plus a reference (ABX) to judge preference or closeness to the reference. They are very common in evaluating fine-grained or zero-shot TTS. Appendix A.4 details the metric computations, while Table 4 summarizes the most commonly used ones.

4.2.2 Model-based Evaluation

Model-based evaluation is also an emerging technique, e.g., automatic MOS evaluation (Lian et al., 2025) and GPT-based evaluation (Rong et al., 2025b,a). To evaluate the controllability of existing TTS models, we designed a pipeline using Google Gemini to assess synthesized speech along three dimensions, i.e., instruction following, naturalness, and expressiveness, which are not well captured by traditional metrics. Details of this evaluation are provided in Appendix A.5.

5 Challenges and Future Directions

5.1 Challenges

Fine-Grained Attribute Control. Emotion and other vocal traits are often intertwined and span

multiple granularities, making fine-grained control especially difficult. This requires high-resolution annotations and advanced models capable of capturing subtle attribute variations. While description-based methods like VoxInstruct (Zhou et al., 2024) allow control via attribute descriptions, precisely targeting a specific granularity or enabling multi-scale, fine-grained control remains a big challenge.

Feature Disentanglement. Fully controllable TTS requires effective feature disentanglement, yet extracting meaningful and independent speech attributes is challenging due to their interdependence and context sensitivity. For instance, modifying pitch can also affect emotion and naturalness. To address this, prior work (An et al., 2022; Wang et al., 2023b) leverages pre-trained models on tasks like emotion classification and adversarial training to guide feature separation. However, designing disentanglement methods for more subtle prosodic attributes, such as sarcasm, remains an open challenge and merits further research.

Scarcity of Datasets. Effective control requires training data that spans a wide range of styles, emotions, accents, and prosodic patterns. Large-scale datasets like GigaSpeech (Chen et al., 2021) and TextrolSpeech (Ji et al., 2024b) exist, but lack the content and scenario diversity, e.g., comedies, thrillers, and cartoons. Fine-grained, attribute-specific annotations are another bottleneck. Manual labeling is expensive, laborious, requires expertise, and is often inconsistent, especially for subjective traits like emotion. Most datasets offer only coarse labels (e.g., gender, age, emotions). While datasets like SpeechCraft (Jin et al., 2024) and Parler-TTS (Lyth and King, 2024) include textual descriptions, none provide annotations across varying conditions within the same speaker.

5.2 Future Directions

Instruction-Guided Fine-Grained Speech Synthesis and Editing. Natural language-driven control of fine-grained speech attributes remains underexplored. Most existing methods support only a limited set of controllable attributes. While models like VoxInstruct (Zhou et al., 2024) and CosyVoice (Du et al., 2024) show promise in controlling emotion, timbre, and style, they often produce speech that deviates from user intent, requiring multiple synthesis attempts. Similarly, speech editing methods (Tae et al., 2022; Tan et al., 2021a) typically rely on conditional models with fixed inputs, offering limited flexibility for fine-grained, instruction-guided modifications. Thus, developing disentangled representations that support precise control through user instructions is promising.

Expressive Multimodal Speech Synthesis. Synthesizing speech from multimodal inputs such as texts, images, and videos has broad industrial applications in storytelling, film, and gaming. While prior work (Goto et al., 2020; Lu et al., 2022; Rong and Liu, 2025) explores this direction, current methods struggle to effectively extract and utilize rich multimodal information. Generating expressive, engaging speech for complex visual content remains a promising area for future research.

Zero-shot Long Speech and Conversational Synthesis with Emotion Consistency. Zero-shot TTS enables voice cloning and style transfer without fine-tuning (Wang et al., 2025b; Chen et al., 2024d; Du et al., 2024), yet struggles with generating long, content-emotion consistent speech due to limited reference input. Overcoming this challenge is key to advancing long speech synthesis. Besides, conversational TTS, often cascaded and context-unaware, produces robotic and unexpressive speech. Recent advances leverage LLMs and discrete speech tokens (Fang et al., 2025; Zhang et al., 2023b), but context-aware, emotionally rich conversational TTS remains underexplored.

Large-Scale Dataset Generation. Dataset construction is critical for both fine-grained control and editing tasks. Researchers can leverage pre-trained speech analysis models to annotate attributes like pitch, energy, emotion, gender, and age. From these annotations, tools like ChatGPT can generate diverse natural language descriptions of speech characteristics. For speech editing, pre-trained models can assist with tasks such as word substitution, pitch adjustment, and emotion conversion, while ChatGPT can provide varied and semantically rich editing instructions for training and evaluation. In addition, multi-agent systems can also be utilized to generate long-form and diverse speech content.

6 Conclusion

This survey provides a comprehensive review of controllable TTS methods, covering model architectures, control strategies, and feature representations. We also summarize commonly used datasets and evaluation metrics, discuss major challenges, and highlight promising future directions. To the best of our knowledge, this is the first comprehensive survey dedicated to controllable TTS.

7 Limitations

We acknowledge several limitations that may affect the completeness of our survey. First, this survey does not explore the interactions between controllable attributes. Most existing studies focus on modeling each factor, such as emotion, speaker identity, or prosody, as an independent variable. However, understanding how these attributes influence one another during synthesis could lead to more effective and flexible control strategies. Second, we do not address the efficiency of current controllable TTS systems. It is important to note that approaches guided by descriptions or instructions often involve considerable computational cost, largely due to their reliance on large language model-based codecs and complex cross-modal architectures. Third, we have not discussed the broader societal implications of controllable TTS, such as the risks associated with deepfake generation or adversarial attacks. Finally, this survey does not cover related research areas, including speech enhancement, speech separation, speech pretraining, and speech-to-speech translation, which may offer valuable insights or complementary techniques. Overcoming these limitations presents important opportunities for future research to deepen our understanding and improve the design of controllable TTS systems.

8 Ethics Statements

The literature search and review were conducted using sources such as Google Scholar, arXiv, DBLP, Scopus, and ChatGPT. All referenced papers included in this survey were thoroughly read, analyzed, and categorized by the authors. Portions of the original content in this survey were paraphrased and refined with the assistance of ChatGPT.

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A Appendix

A.1 The History of Controllable TTS

Controllable TTS aims to steer various aspects of synthesized speech, including pitch, energy, speed, prosody, timbre, emotion, gender, and speaking style. This subsection briefly reviews its development, from early methods to recent advancements.

Early Methods. Early controllable TTS systems were primarily based on rule-based, concatenative, and statistical approaches. Rule-based systems, such as formant synthesis (Rabiner, 1968; Allen et al., 1987; Purcell and Munhall, 2006), used handcrafted rules to adjust acoustic parameters like pitch and duration, enabling basic prosody control. Concatenative systems (Hunt and Black, 1996; Wouters and Macon, 2001; Bulut et al., 2002) generated speech by stitching pre-recorded speech units together, allowing prosody modifications through pitch and timing adjustments. Later, Hidden Markov model (HMM)-based statistical methods (Nose et al., 2007; Ling et al., 2009; Tokuda et al., 2000) modeled the relationship between linguistic features and acoustic outputs, offering

Control Strategy	Core Idea	Key Features	Pros & Cons
Style Tagging	Control specific attributes using predefined tags.	Direct control over attributes like emotion and pitch.	**Pros**: Simple to implement. **Cons**: Limited expressive diversity, cannot achieve fine-grained or combined control.
Reference Speech Prompts	Use a short audio clip as a reference for style.	Enables zero-shot TTS and voice cloning; separates timbre and style.	**Pros**: High degree of personalization, more flexible control. **Cons**: Relies on high-quality reference audio, less intuitive control dimensions.
Natural Language Descriptions	Describe desired voice characteristics in text.	User-friendly control via natural language (e.g., "speak calmly").	**Pros**: High interpretability, user-friendly. **Cons**: Limited freedom and accuracy of description, potential for model misunderstanding.
Instruction-Guided Control	Use LLMs to interpret free-form instructions.	Combines content and style instructions; can generate paralinguistic sounds.	**Pros**: Extremely high control precision and freedom, understands complex instructions. **Cons**: Strong dependency on LLMs, high system complexity.

Table 2: Summary of the pros and cons of each control strategy.

Dataset	Hours (at least)	#Speakers (at least)	Labels											Lang	Release Time
			Pit.	Ene.	Spe.	Age	Gen.	Emo.	Emp.	Acc.	Top.	Des.	Dia.		
IEMOCAP (Busso et al., 2008)	12	10	✓	✓	✓		✓	✓						en	2008
RECOLA (Ringeval et al., 2013)	3.8	46						✓						fr	2013
RAVDESS (Livingstone and Russo, 2018)	/	24				✓		✓						en	2018
CMU-MOSEI (Bagher Zadeh et al., 2018)	65	1,000						✓						en	2018
Taskmaster-1 (Byrne et al., 2019)	/	/											✓	en	2019
AISHELL-3 (Shi et al., 2021)	85	218				✓	✓			✓				zh	2020
Common Voice (Ardila et al., 2020)	2,500	50,000				✓	✓			✓				multi	2020
ESD (Zhou et al., 2022)	29	10						✓						en,zh	2021
GigaSpeech (Chen et al., 2021)	10,000	/									✓			en	2021
WenetSpeech (Zhang et al., 2022)	10,000	/									✓			zh	2021
PromptSpeech (Guo et al., 2023)	/	/	✓	✓	✓		✓					✓		en	2022
MagicData-RAMC (Yang et al., 2022)	180	663									✓		✓	zh	2022
DailyTalk (Lee et al., 2023a)	20	2						✓			✓		✓	en	2023
TextrolSpeech (Ji et al., 2024b)	330	1,324	✓	✓	✓		✓	✓				✓		en	2023
CLESC (Toloka, 2024)	<1	/	✓	✓	✓			✓						en	2024
VccmDataset (Ji et al., 2024c)	330	1,324	✓	✓	✓		✓	✓				✓		en	2024
MSceneSpeech (Yang et al., 2024d)	13	13									✓			zh	2024
Parler-TTS (Lyth and King, 2024)	50,000	/	✓		✓		✓	✓		✓		✓		en	2024
SpeechCraft (Jin et al., 2024)	2,391	3,200	✓	✓	✓	✓	✓	✓	✓		✓	✓		en,zh	2024

Abbreviations: Pit(ch), Ene(rgy)=volume, Spe(ed), Gen(der), Emo(tion), Emp(hasis), Acc(ent), Top(ic), Des(cription), Dia(logue).

Table 3: A summary of publicly available speech datasets for controllable TTS.

Metric	Type	Eval Target	GT Required
MCD (Kominek et al., 2008)↓	Objective	Acoustic similarity	✓
FDSD (Bińkowski et al., 2020)↓	Objective	Acoustic similarity	✓
WER (Wikipedia, 2024)↓	Objective	Intelligibility	✓
Cosine (Desplanques et al., 2020)↓	Objective	Speaker similarity	✓
PESQ (Rix et al., 2001)↑	Objective	Perceptual quality	✓
MOS (Wikipedia, 2025)↑	Subjective	Preference	
CMOS (Loizou, 2011)↑	Subjective	Preference	
AB Test	Subjective	Preference	
ABX Test	Subjective	Perceptual similarity	✓

GT: Ground truth, ↓: Lower is better, ↑: Higher is better.

Table 4: Widely used evaluation metrics.

greater flexibility in controlling prosody and speaking rate. These systems also introduced speaker adaptation (Yamagishi et al., 2009) and limited emotional control (Yamagishi et al., 2003), and require less storage and provide smoother transitions than concatenative methods.

Neural Synthesis. The emergence of deep learning revolutionized TTS, leading to the development of neural model-based systems capable of producing more natural, expressive, and controllable speech. Early models like WaveNet (Van Den Oord et al., 2016) and Tacotron (Wang et al., 2017) demonstrated the potential for prosody control through explicit conditioning (Shen et al., 2018; Ren et al., 2021b). Neural TTS further enhanced speaker control through speaker embeddings and adaptation techniques (Fan et al., 2015; Casanova et al., 2022), while advances in emotional modeling (Lei et al., 2022; Um et al., 2020) enabled the synthesis of speech with specific affective tones. Recent models have also achieved manipulation of timbre (Wang et al., 2025b; Shen et al., 2024) and style (Li et al., 2025b; Huang et al., 2022b), fostering the research in zero-shot TTS and voice cloning (Cooper et al., 2020). In addition, methods for fine-grained content control (Peng et al., 2024b; Tan et al., 2021a) have made it possible to emphasize or edit specific

words in synthesized speech.

LLM-based Synthesis. More recently, LLM-based approaches have further advanced controllable TTS. Leveraging models like BERT (Devlin et al., 2019), GPT (Brown et al., 2020), T5 (Raffel et al., 2020), and PaLM (Chowdhery et al., 2023), LLMs bring superior context modeling and intuitive control to speech synthesis (Guo et al., 2023; Zhou et al., 2024). By interpreting natural language prompts, such as describing a speaker's emotion, age, or style, LLMs can infer nuanced attributes and steer the generation process accordingly. This enables dynamic, fine-grained control over prosody, emotion, style, and speaker identity (Yang et al., 2024b; Gao et al., 2024), marking a big step toward more flexible and intuitive TTS systems.

A.2 Overview of the TTS Pipeline

In this section, we provide an overview of the general pipeline that supports controllable TTS technologies. Fig. 5 depicts the general pipeline of controllable TTS, containing various model architectures and feature representations.

A TTS pipeline generally contains three key components, i.e., linguistic analyzer, acoustic model, and speech vocoder, where a conditional input, e.g., prompts, can be processed for controllable speech synthesis. *Linguistic analyzer* aims to extract linguistic features, e.g., phoneme duration and position, syllable stress, and utterance level, from the input text, which is a necessary step in HMM-based methods (Yoshimura et al., 1999; Tokuda et al., 2000) and a few neural model-based methods (Zen et al., 2013; Fan et al., 2014), but is time-consuming and error-prone. *Acoustic model* is a parametric or neural model that predicts the acoustic features from the input texts. Modern neural acoustic models like Tacotron (Wang et al., 2017) and later works (Ren et al., 2019, 2021b; Jeong et al., 2021) directly take character (Chen et al., 2015) or word embeddings (Almeida and Xexéo, 2019) as input, which is much more efficient than previous methods. *Speech vocoder* is the last component that converts the intermediate acoustic features into a waveform that can be played back. This step bridges the gap between the acoustic features and the actual sounds produced, helping to generate high-quality, natural-sounding speech (Van Den Oord et al., 2016; Kong et al., 2020). Besides, some end-to-end methods use a single model to encode the input and decode the speech waveforms without generating intermediate features like mel-spectrograms. One can refer to Tan et al. (2021b) for a more comprehensive and detailed review of acoustic models and vocoders.

A.3 Speech Quantization vs. Tokenization.

It is worth noting that quantization and tokenization serve distinct purposes in speech processing. Quantization is primarily used for high-fidelity compression, reducing the precision of numerical representations (e.g., from 32-bit floating point to 8-bit integers) while preserving model performance. In speech synthesis, quantization is often used in waveform generation (e.g., codec-based approaches like EnCodec (Défossez et al., 2023b)) and neural vocoders to compress audio signals without significant loss of perceptual quality. Tokenization, on the other hand, is a discretization process that segments continuous data into meaningful units. In speech tasks, tokenization extracts semantically relevant representations such as phonemes, characters, or learned speech units (e.g., HuBERT (Hsu et al., 2021) and Wav2Vec 2.0 (Baevski et al., 2020b)). This makes tokenization particularly suitable for speech-to-text (ASR), TTS, and multimodal NLP tasks, where aligning speech with textual information is crucial. Tokenization also facilitates training language models on speech data by enabling linguistic or learned unit-based processing rather than raw audio waveform modeling. Table 5 in Appendix A.3 summarizes popular open-source speech quantization and tokenization methods. Table 1 summarizes the acoustic features of representative methods.

A.4 Evaluation Metric Computations

The performance of controllable TTS often requires objective and subjective evaluation. We introduce common evaluation metrics in this subsection.

Objective Evaluation Metrics. Objective metrics offer automated and reproducible evaluations. Mel Cepstral Distortion (MCD) (Kominek et al., 2008) measures the spectral distance between synthesized and reference speech, reflecting how closely the generated audio matches the target in terms of acoustic features. A lower MCD value indicates a higher similarity between synthesized and reference speech, meaning better speech synthesis quality. Typically, an MCD value below 4 suggests good quality, while values above 6 may indicate significant distortion. The MCD is computed as

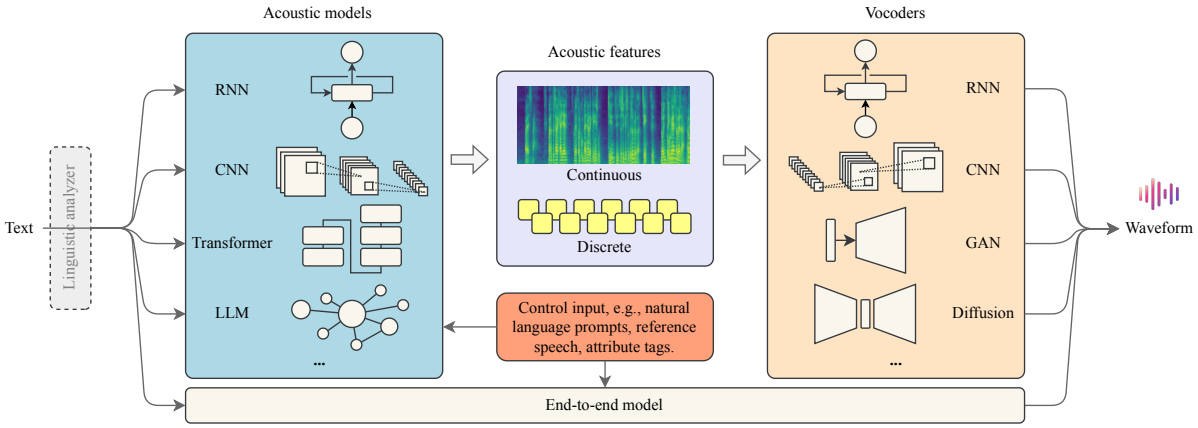


Figure 5: General pipeline of controllable TTS from the perspective of network structure. Linguistic analysis is necessary for parametric and a few neural methods but is no longer needed for most modern neural methods. In this paper, we only review neural model-based controllable TTS methods and do not investigate acoustic features (e.g., MFCC (Fukada et al., 1992), LSP (Itakura, 1975), F0 (Kawahara et al., 1999)) used in early TTS methods.

Method	Modeling	Code	Year
VQ-Wav2Vec (Baevski et al., 2020a)	SSCP	https://github.com/facebookresearch/fairseq/tree/main/examples/wav2vec#vq-wav2vec	2019
Wav2Vec 2.0 (Baevski et al., 2020b)	SSCP	https://github.com/facebookresearch/fairseq/tree/main/examples/wav2vec	2019
HuBERT (Hsu et al., 2021)	SSCP	https://github.com/facebookresearch/fairseq/tree/main/examples/hubert	2021
Whisper Encoder (Radford et al., 2023)	SSCP	https://github.com/openai/whisper	2022
Data2vec (Baevski et al., 2022)	SSCP	https://github.com/facebookresearch/fairseq/tree/main/examples/data2vec	2022
W2v-BERT 2.0 (Barrault et al., 2023)	SSCP	https://huggingface.co/facebook/w2v-bert-2.0	2023
SoundStream (Zeghidour et al., 2021)	RVQ-GAN	https://github.com/wesbz/SoundStream	2021
Encodec (Défossez et al., 2023b)	RVQ-GAN	https://github.com/facebookresearch/encodec	2022
HiFi-Codec (Yang et al., 2023a)	RVQ-GAN	https://github.com/yangdongchao/AcademiCodec	2023
SpeechTokenizer (Zhang et al., 2024)	RVQ-GAN	https://github.com/ZhangXInFD/SpeechTokenizer	2023
Descript Audio Codec (Kumar et al., 2024)	RVQ-GAN	https://github.com/descriptinc/descript-audio-codec	2023
Mimi Codec (Défossez et al., 2024)	RVQ-GAN	https://github.com/kyutai-labs/moshi	2024
WavTokenizer (Ji et al., 2025)	VQ-GAN	https://github.com/jishengpeng/WavTokenizer	2024

SSCP: Self-supervised context (token) prediction, RVQ: Residual vector quantization (Zeghidour et al., 2021).

Table 5: Popular open-source speech quantization and tokenization methods.

follows:

$$MCD = \frac{10}{\ln 10} \cdot \sqrt{2 \sum_{d=1}^{D} (c_d^{(syn)} - c_d^{(ref)})^2}, \quad (3)$$

where $c_d^{(syn)}$ represents the d-th Mel Cepstral Coefficient (MCC) of the synthesized speech, $c_d^{(ref)}$ represents the d-th MCC of the reference speech, D is the number of MCC, and $\frac{10}{\ln 10} \approx 4.342$ is a constant factor that converts the logarithm to a decibel scale.

Fréchet DeepSpeech Distance (FDSD) (Bińkowski et al., 2020) is another metric designed to evaluate the quality and naturalness of synthesized speech. It is inspired by the Fréchet Inception Distance (FID) (Heusel et al., 2017) used in image generation but adapted to speech by leveraging a deep speech recognition model. FDSD measures the statistical distance between the distributions of real (reference) and synthesized speech in the feature space of a pretrained speech recognition model, such as Deep Speech (Hannun et al., 2014). By comparing the mean and covariance of the extracted feature representations, FDSD provides a perceptually relevant assessment of speech synthesis quality. A lower FDSD means the synthesized speech is more similar to real speech. FDSD can be computed as:

$$FDSD = ||\mu_s - \mu_r||^2 + \text{Tr}(\Sigma_s + \Sigma_r - 2(\Sigma_s \Sigma_r)^{1/2}), \quad (4)$$

where μ_s and Σ_s are the mean and covariance of the embeddings from the synthesized speech, μ_r and Σ_r are the mean and covariance of the embeddings from the real (reference) speech, $||\mu_s - \mu_r||^2$ represents the squared Euclidean distance between the means, $\text{Tr}(\cdot)$ denotes the trace of a matrix, and $(\Sigma_s \Sigma_r)^{1/2}$ is the geometric mean of the covariance matrices.

For intelligibility, the Word Error Rate (WER) (Wikipedia, 2024) is used. It measures the difference between the recognized transcript and the reference transcript by computing the number

of errors made in the transcription process. WER is computed as:

$$WER = \frac{S + D + I}{N}, \tag{5}$$

where S is the number of substitutions (wrong word in place of the correct word), D is the number of deletions (missed words), I is the number of insertions (extra words added), and N is the total number of words in the reference transcript.

Cosine similarity (on speaker embeddings) measures similarity between the speaker embeddings of synthesized and reference speech. It can be used to evaluate zero-shot TTS (voice cloning) methods, where higher values indicate better speaker similarity. Given two speaker embeddings, $\mathbf{e_1}$ and $\mathbf{e_2}$, their cosine similarity is defined as:

$$CosSim(\mathbf{e_1}, \mathbf{e_2}) = \frac{\mathbf{e_1} \cdot \mathbf{e_2}}{\|\mathbf{e_1}\|\|\mathbf{e_2}\|}, \tag{6}$$

where speaker embeddings can be extracted from a pre-trained speaker embedding model (e.g., ECAPA-TDNN (Desplanques et al., 2020) and x-vectors (Snyder et al., 2018)).

Perceptual Evaluation of Speech Quality (PESQ) (Rix et al., 2001) is another objective metric designed to evaluate speech quality by comparing degraded audio with a clean reference. It is widely used in telecommunications and speech synthesis. PESQ models human auditory perception, producing a score in the range $[-0.5, -4.5]$ that reflects intelligibility and distortion under various conditions, including noise or compression. PESQ involves complex perceptual modeling, its core components can be summarized as:

$$PESQ = a_0 + a_1 \cdot D_{frame} + a_2 \cdot D_{time}, \tag{7}$$

where D_{frame} is the frame-by-frame perceptual distortion, D_{time} is the time-domain distortion, and a_0, a_1, a_2 are regression coefficients. One can refer to (Rix et al., 2001) for details.

Signal-to-Noise Ratio (SNR) measures the ratio of signal power to noise power. A higher SNR indicates a cleaner signal with less noise, while a lower SNR suggests that noise is dominating the signal. However, in TTS, noise can come from different sources, such as artifacts from vocoders, neural network distortions, or background noise in dataset recordings. A direct computation of SNR in TTS requires a reference clean speech signal ($x[n]$), a synthesized (or noisy) speech signal ($y[n]$), and extracting the noise component ($e[n] = y[n] - x[n]$) from the synthesized signal. The SNR for TTS systems can be computed as:

$$SNR = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right), \tag{8}$$

where $P_{\text{signal}} = \frac{1}{N} \sum_{n=1}^{N} x[n]^2$ and $P_{\text{noise}} = \frac{1}{N} \sum_{n=1}^{N} e[n]^2$.

Subjective Evaluation Metrics. The Mean Opinion Score (MOS) (Wikipedia, 2025) is the most commonly used subjective metric. In MOS evaluations, listeners rate various aspects, such as naturalness, expressiveness, quality, intelligibility, et al., of synthesized speech on a scale from 1 to 5, where higher scores indicate better quality. MOS captures human perception effectively, but is expensive for large-scale evaluations.

Comparison Mean Opinion Score (CMOS) (Loizou, 2011) further evaluates relative quality differences between two TTS audio samples. Participants listen to paired samples and rate their preference on a scale (e.g., -3 to +3, where negative values favor the first sample). CMOS is used to measure subtle improvements in TTS systems, complementing absolute MOS ratings. MOS and CMOS scores are computed as the average scores across all listeners:

$$MOS/CMOS = \frac{1}{N} \sum_{i=1}^{N} s_i, \tag{9}$$

where s_i is the score given by the i-th listener, and N is the number of listeners.

AB and ABX tests are also popular in evaluating TTS methods. An AB test involves presenting two versions of a synthesized speech (from different TTS models) to human listeners and asking them to choose which they prefer. The goal is to assess which model produces better-sounding speech based on certain criteria, such as naturalness, intelligibility, or clarity. In an ABX test, listeners compare two synthesized speech samples to a reference speech sample and determine which one is closer in terms of timbre, prosody, emotion, and other relevant features. ABX tests are widely used in evaluating zero-shot TTS methods. The AB/ABX test score for a model m is:

$$Score_{AB}/Score_{ABX} = \frac{N_m}{N}, \tag{10}$$

where N_m represents the number of listeners who prefer the speech synthesized by model m, and N denotes the total number of listeners.

Table 4 summarizes widely used metrics for TTS.

A.5 A Google Gemini-Based Experimental Evaluation of TTS Controllability

We designed an evaluation pipeline using Gemini to assess synthesized speech in terms of **instruction following**, **naturalness**, and **expressiveness**, because these dimensions are not well captured by traditional metrics. Conventional scores like MCD, WER, PESQ, speaker similarity, and MOS/CMOS are excluded, as our goal is to explore the feasibility of using multimodal large language models (MLLMs) as subjective evaluators.

A.5.1 Implementation details

Models. Due to time constraints, we only evaluate a total of 10 models: 8 open-source systems (F5-TTS, CosyVoice, CosyVoice2, Vevo, Spark-TTS, MaskGCT, PromptTTS, and VoxInstruct) and 2 commercial TTS systems (ElevenLabs and MiniMax TTS).

Tasks. Zero-shot TTS and description-based synthesis. For each model, we synthesize 20 speech samples (10 in English and 10 in Chinese) for each task.

Dataset. For zero-shot TTS, we sampled 10 English utterances from the MSP-Podcast dataset and 10 Chinese utterances from Emo-Emilia to serve as reference speech prompts. For description-based synthesis, we used ChatGPT to generate diverse textual descriptions as shown in Fig. 6.

Metrics Clarification:

- Instruction Following
  - Purpose: To assess how accurately the synthesized audio follows the given instruction regarding speech characteristics such as timing, emphasis, and pacing.
  - Focus: Measures the controllability and fidelity of the model in executing user-specified directives.
- Naturalness
  - Purpose: To evaluate how natural the audio sounds—whether it resembles human speech or exhibits synthetic, robotic qualities.
  - Focus: Measures the perceptual audio quality and realism of the synthesized speech.
- Expressiveness
  - Purpose: To judge the emotional richness and prosodic variation in the audio, such as tone, intensity, and nuance.
  - Focus: Measures the model's ability to convey expressive and emotionally engaging speech.

Gemini Prompts.

The prompt we use for the evaluation is as follows:

SYSTEM_PROMPT: You are a strict quality evaluator for synthesized speech. Given an audio file of a speech sample, its transcript, and an instruction describing the intended speech characteristics, please rate the audio based on the following three aspects, using the defined criteria. Output ONLY a JSON dictionary with the keys instruction_following, naturalness, and expressiveness, each assigned an integer value from 1 to 5. The evaluation rubrics are as follows:

1. **Instruction Following (1–5):**
   - 1 point: The audio completely ignores the instruction; it does not follow the intended timing, emphasis, or pacing.
   - 2 points: It loosely follows the instructions but misses key elements or timing in parts.
   - 3 points: Generally follows the instruction with minor lapses in emphasis or pacing.
   - 4 points: Clearly follows the instruction with only slight deviations.
   - 5 points: Perfectly follows every aspect of the instruction with clear emphasis and precise pacing.
2. **Naturalness (1–5):**
   - 1 point: The audio sounds fully synthetic or robotic; extremely unnatural.
   - 2 points: Noticeably synthetic; some unnatural artifacts remain.
   - 3 points: Moderately natural with occasional synthetic artifacts.
   - 4 points: Largely natural sounding with minor imperfections.
   - 5 points: Completely natural; indistinguishable from a human recording.
3. **Expressiveness (1–5):**

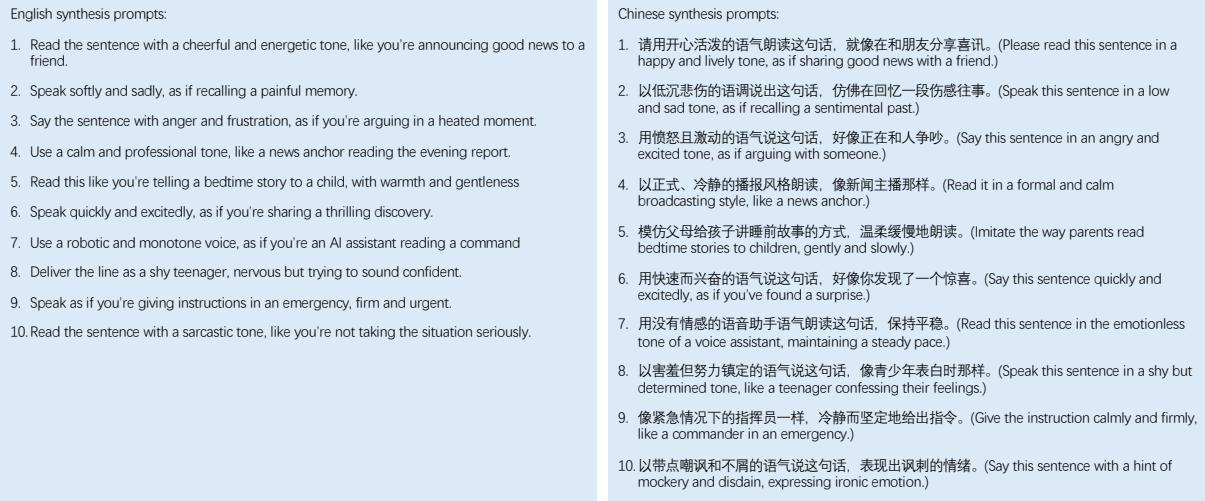


Figure 6: Textual descriptions generated by ChatGPT

- 1 point: The audio is flat and monotone; no emotional variation.
- 2 points: Minimal expressiveness; emotions are weak or inconsistent.
- 3 points: Reasonably expressive with some highlights, but could be stronger.
- 4 points: Clearly expressive with only slight under- or over-emphasis.
- 5 points: Exceptionally expressive; full emotional richness and nuance.

USER_PROMPT: The synthesized speech is {audio}. The transcript of the audio is: "{transcript}". The instruction for the audio is: "{instruction}".

A.5.2 Results: Model-level Performance Comparison

As shown in Table 6, in the zero-shot setting, among the six models, Vevo performs best in both naturalness (4.43±0.55) and expressiveness (4.32±0.75), indicating strong general quality without explicit guidance. CosyVoice, CosyVoice 2, and F5-TTS follow closely with similar scores (4.2), while SparkTTS and MaskGCT lag behind, especially in naturalness.

In the instruction-based setting, all models show a clear improvement across all metrics. CosyVoice achieves the highest overall scores, with instruction following at 4.81±0.28, naturalness at 4.92±0.24, and expressiveness at 4.78±0.29. Other strong performers include MiniMax TTS and EmoVoice, both exceeding 4.6 in most dimensions. Even the lowest-scoring instruction-based method (VoxInstruct) outperforms the best zero-shot model in every aspect.

A.5.3 Results: The Reliability of Multimodal LLM-based Evaluation

We also compare the proposed metrics with existing automated evaluation methods, namely NISQA (Mittag et al., 2021) and UTMOS (Saeki et al., 2022). Specifically, we use 96 synthesized samples to compute the Pearson correlation coefficients between the predicted scores from each method and human ratings, aiming to assess how well each method aligns with human perception.

As shown in Table 7, although the absolute Pearson correlation coefficients of our method are relatively modest, our approach consistently outperforms both NISQA and UTMOS across all three evaluation dimensions: instruction following, naturalness, and expressiveness.

These results suggest that existing automated metrics like NISQA and UTMOS, which are primarily designed for general speech quality assessment, may not capture nuanced attributes such as speaker intent or expressive delivery in controllable TTS tasks.

In contrast, our metric, tailored for instruction-based synthesis evaluation, better reflects human judgments, particularly in aspects beyond raw audio quality. This supports the need for task-specific evaluation frameworks when benchmarking modern controllable TTS systems.

A.5.4 Conclusion

To some extent, the proposed MLLM-based evaluation pipeline is able to predict human-aligned scores for instruction following, naturalness, and expressiveness. We also find that it offers promising potential for automated evaluation of control-

Task	Method	Instruction Following (Gemeni 2.5 Flash / Human)	Naturalness (Gemeni 2.5 Flash / Human)	Expressiveness (Gemeni 2.5 Flash / Human)
Zero-shot	F5-TTS	-	4.27±0.87	4.21±0.78
	CosyVoice	-	4.20±0.82	4.17±0.83
	CosyVoice 2	-	4.25±0.58	4.20±0.63
	Vevo	-	4.43±0.55	4.32±0.75
	SparkTTS	-	3.68±0.80	3.83±0.79
	MaskGCT	-	3.91±0.88	4.08±0.81
Instruction-based	CosyVoice	4.81±0.28	4.92±0.24	4.78±0.29
	CosyVoice 2	4.61±0.49	4.85±0.31	4.63±0.52
	EmoVoice	4.67±0.44	4.80±0.36	4.67±0.44
	VoxInstruct	4.45±0.50	4.83±0.32	4.50±0.52
	ElevenLabs	4.52±0.67	4.85±0.31	4.63±0.57
	MiniMax TTS	4.67±0.36	4.87±0.27	4.63±0.44

Table 6: The evaluation of the controllability of open-source and commercial TTS systems.

	Instruction Following	Naturalness	Expressiveness
NISQA (Mittag et al., 2021)	-	0.01	-0.03
UTMOS (Saeki et al., 2022)	-	-0.10	-0.17
Ours	0.12	0.17	0.14

Table 7: The alignment between model-based evaluation and human preference.

lable TTS.

In future work, we plan to enhance our survey by designing a more robust and reliable MLLM-based evaluation framework and conducting a comprehensive benchmark of existing controllable TTS methods.